

Лабораторная работа №1

Подготовка лабораторного стенда

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17 декабря 2025

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Информация

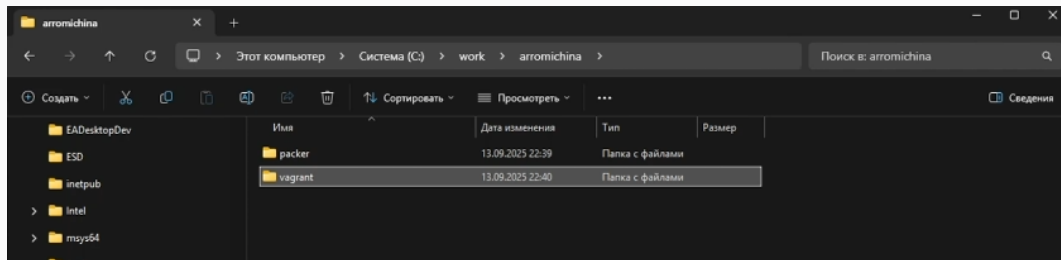
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Вводная часть

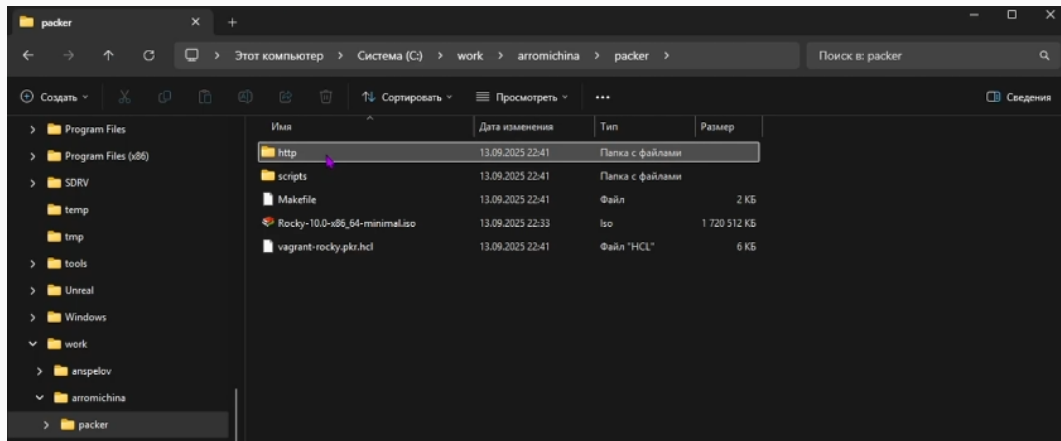
- Целью данной работы является приобретение практических навыков установки Rocky Linux на виртуальную машину с помощью инструмента Vagrant.

Основная часть

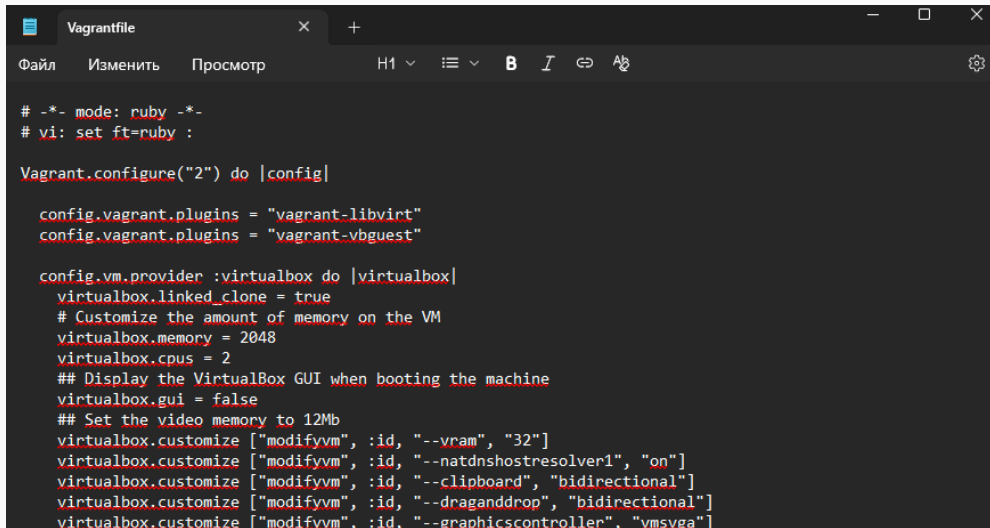
- Создание каталога для проекта.



- Размещение образа варианта операционной системы Rocky Linux в рабочем каталоге.



- Проверка конфигурационного файла Vagrantfile.

A screenshot of a code editor window titled 'Vagrantfile'. The editor has a dark theme and shows a Ruby configuration file. The code defines Vagrant plugins (libvirt and vbguest) and configures a VirtualBox VM with specific settings for memory, CPUs, GUI, and video. The code is as follows:

```
# -*- mode: ruby -*-
# vi: set ft=ruby :

Vagrant.configure("2") do |config|

  config.vagrant.plugins = ["vagrant-libvirt"]
  config.vagrant.plugins = ["vagrant-vbguest"]

  config.vm.provider :virtualbox do |virtualbox|
    virtualbox.linked_clone = true
    # Customize the amount of memory on the VM
    virtualbox.memory = 2048
    virtualbox.cpus = 2
    ## Display the VirtualBox GUI when booting the machine
    virtualbox.gui = false
    ## Set the video memory to 12Mb
    virtualbox.customize ["modifyvm", :id, "--vram", "32"]
    virtualbox.customize ["modifyvm", :id, "--natdnshostresolver1", "on"]
    virtualbox.customize ["modifyvm", :id, "--clipboard", "bidirectional"]
    virtualbox.customize ["modifyvm", :id, "--draganddrop", "bidirectional"]
    virtualbox.customize ["modifyvm", :id, "--graphicscontroller", "vmsvga"]
```

Установка окружения виртуальной машины

- Ввод необходимых команд.

```
C:\work\arrowichina\packer>packer.exe build vagrant-rocky.pkr.hcl
virtualbox-iso.rockylinux: output will be in this color.
qemu.rockylinux: output will be in this color.

Build 'qemu.rockylinux' errored after 41 milliseconds 4 microseconds: Failed creating Qemu driver: exec: "qemu-system-x86_64": executable file not found in %PATH%
==> virtualbox-iso.rockylinux: Retrieving Guest additions
==> virtualbox-iso.rockylinux: Trying C:\Program Files\Oracle\VirtualBox\VBGuestAdditions.iso
==> virtualbox-iso.rockylinux: Trying file://C:/Program%20Files/Oracle/VirtualBox/VBoxGuestAdditions.iso
==> virtualbox-iso.rockylinux: file://C:/Program%20Files/Oracle/VirtualBox/VBoxGuestAdditions.iso => C:/Program Files/Oracle/VirtualBox/VBoxGuestAdditions.iso
==> virtualbox-iso.rockylinux: Retrieving ISO
==> virtualbox-iso.rockylinux: Trying Rocky-10.0-x86_64-minimal.iso
==> virtualbox-iso.rockylinux: Trying Rocky-10.0-x86_64-minimal.iso?checksum=sha256%3Ade75c2f7cc566ea964017a1e94883913f066c4ebeb1d356964e398ed76cadd12
==> virtualbox-iso.rockylinux: Rocky-10.0-x86_64-minimal.iso?checksum=sha256%3Ade75c2f7cc566ea964017a1e94883913f066c4ebeb1d356964e398ed76cadd12 => C:/work/arrowichina/packer/Rocky-10.0-x86_64-minimal.iso
==> virtualbox-iso.rockylinux: Starting HTTP server on port 8285
==> virtualbox-iso.rockylinux: Creating virtual machine...
==> virtualbox-iso.rockylinux: Creating hard drive output-rockylinux10-virtualbox\rockylinux10-virtualbox.vdi with size 61440 MiB...
==> virtualbox-iso.rockylinux: Mounting ISOs...
==> virtualbox-iso.rockylinux: Mounting boot ISO...
==> virtualbox-iso.rockylinux: Creating forwarded port mapping for communicator (SSH, WinRM, etc) (host port 3117)
==> virtualbox-iso.rockylinux: Executing custom VBoxManage commands...
==> virtualbox-iso.rockylinux: Executing: modifyvm rockylinux10-virtualbox --memory 2048
==> virtualbox-iso.rockylinux: Executing: modifyvm rockylinux10-virtualbox --cpus 2
==> virtualbox-iso.rockylinux: Executing: modifyvm rockylinux10-virtualbox --nat-localhostreachabl on
==> virtualbox-iso.rockylinux: Executing: modifyvm rockylinux10-virtualbox --firmware EFI
==> virtualbox-iso.rockylinux: Executing: modifyvm rockylinux10-virtualbox --vrde on
==> virtualbox-iso.rockylinux: Executing: modifyvm rockylinux10-virtualbox --vrdeport 3390
==> virtualbox-iso.rockylinux: Starting the virtual machine...
==> virtualbox-iso.rockylinux: The VM will be run headless, without a GUI. If you want to
==> virtualbox-iso.rockylinux: view the screen of the VM, connect via VRDP without a password to
==> virtualbox-iso.rockylinux: rdp://127.0.0.1:5981
==> virtualbox-iso.rockylinux: Waiting 10s for boot...
==> virtualbox-iso.rockylinux: Typing the boot command...
==> virtualbox-iso.rockylinux: Using SSH communicator to connect: 127.0.0.1
==> virtualbox-iso.rockylinux: Waiting for SSH to become available...
```

- Регистрации образа виртуальной машины в vagrant.

```
C:\work\arromichina\packer>vagrant box add rockylinux10 vagrant-virtualbox-rockylinux10-x86_64.box
==> box: Box file was not detected as metadata. Adding it directly...
==> box: Adding box 'rockylinux10' (v0) for provider: (amd64)
      box: Unpacking necessary files from: file:///C:/work/arromichina/packer/vagrant-virtualbox-rockylinux10-x86_64.box
      box:
The box you're attempting to add already exists. Remove it before
adding it again or add it with the '--force' flag.
```

Работа с окружением виртуальной машины

- Запуск виртуальной машины Server.

```
C:\work\arromichina\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Cloning VM...
==> server: Matching MAC address for NAT networking...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Setting the name of the VM: vagrant_server_1757795233513_34150
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
==> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
```

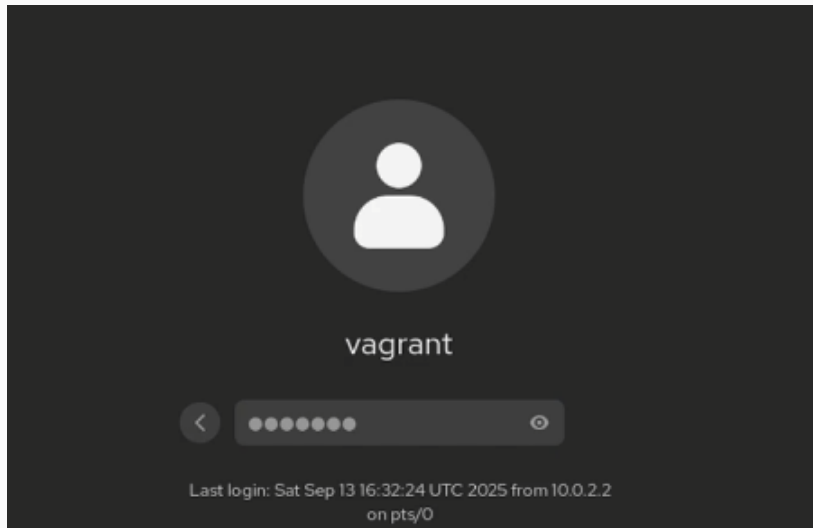
Работа с окружением виртуальной машины

- Запуск виртуальной машины Client.

```
C:\work\arromichina\vagrant>vagrant up client
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Cloning VM...
==> client: Matching MAC address for NAT networking...
==> client: Setting the name of the VM: vagrant_client_1757795476258_8688
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
    client: Adapter 1: nat
    client: Adapter 2: intnet
==> client: Forwarding ports...
    client: 22 (guest) => 2200 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
==> client: Waiting for machine to boot. This may take a few minutes...
    client: SSH address: 127.0.0.1:2200
    client: SSH username: vagrant
    client: SSH auth method: password
==> client: Machine booted and ready!
```

Работа с окружением виртуальной машины

- Вход под пользователем vagrant.



- Подключение к серверу из консоли.

```
C:\work\arromichina\vagrant>vagrant ssh server
==> server: The machine you're attempting to SSH into is configured to use
==> server: password-based authentication. Vagrant can't script entering the
==> server: password for you. If you're prompted for a password, please enter
==> server: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 13 20:36:05 2025
```

- Переход к пользователю arromichina.

```
vagrant@127.0.0.1's password:  
Last login: Sat Sep 13 20:36:05 2025  
vagrant@server:~$ su - arromichina  
Password:  
[arromichina@server.arromichina.net ~]$ |
```

Работа с окружением виртуальной машины

- Проверка на сервере.

```
C:\work\arromichina\vagrant>vagrant ssh client
==> client: The machine you're attempting to SSH into is configured to use
==> client: password-based authentication. Vagrant can't script entering the
==> client: password for you. If you're prompted for a password, please enter
==> client: the same password you have configured in the Vagrantfile.
vagrant@127.0.0.1's password:
Last login: Sat Sep 13 20:35:31 2025
vagrant@client:~$ su - arromichina
Password:
[arromichina@client.arromichina.net ~]$ |
```

- Выключение виртуальных машин.

```
C:\work\arromichina\vagrant>vagrant halt server  
==> server: Attempting graceful shutdown of VM...  
  
C:\work\arromichina\vagrant>vagrant halt client
```

Вывод

- В ходе выполнения лабораторной работы были приобретены практические навыки установки Rocky Linux на виртуальную машину с помощью инструмента Vagrant.